

Table 1. Comparative Program Accomplishments

Milestones	Maximum Pace	Program I	II, III	Low Level
<u>Manned Systems</u>				
Space Station (Earth Orbit)	1975	1976	1977	—
50-Man Space Base (Earth Orbit)	1980	1980	1984	—
100-Man Space Base (Earth Orbit)	1985	1985	1989	—
Lunar Orbiting Station	1976	1978	1981	—
Lunar Surface Base	1978	1980	1983	—
Initial Mars Expedition	1981	1983	II-1986 III-Open	—
<u>Space Transportation System</u>				
Earth-to-Orbit	1975	1976	1977	—
Nuclear Orbit Transfer Stage	1978	1978	1981	—
Space Tug	1976	1978	1981	—
<u>Scientific</u>				
Large Orbiting Observatory	1979	1979	1980	—
High Energy Astron. Capability	1973	1973	1981	1973
Out-of-Ecliptic Survey	1975	1975	1978	1975
Mars—High Resolution Mapping	1977	1977	1981	1977
Venus—Atmospheric Probes	1976	1976	Mid-80s	1976
Multiple Outer Planet "Tours"	1977-79	1977-79	1977-79	1977-79
Asteroid Belt Survey	1975	1975	1981	1975
<u>Applications</u>				
Earliest Oper. Earth Resource System	1975	1975	1976	1975
Demonstration of Direct Broadcast	1978	1978	Mid-80s	1978
Demonstration of Navigation Traffic Control	1974	1974	1976	1974

Space Task Group

These recommendations were not at all what Nixon and most of his advisers had in mind for the post-Apollo space effort; Nixon did not believe that there was public support for continuing a fast-paced space effort, and he gave higher priority to controlling the government budget than to investing in major new space initiatives. It took Nixon almost six months to issue a formal response to the Space Task Group report. In the interim there were cuts to the NASA budget, even after the successes of Apollo 11 and 12. In order to get money to study future programs such as the space station and space shuttle, NASA agreed to end production of the large Saturn V launch vehicle, thereby giving up the U.S. capability to send humans to the Moon and beyond. In this March 7, 1970, statement issued by the White House press office, Nixon set out his views on the principles that should guide decisions on future space efforts. Most significant, he indicated that the space program was no longer to be treated as a high-priority undertaking, but rather as just one of many things that the government would do, and that the space program would have to compete for priority and funding with other government programs. This "Nixon space doctrine" has been the guiding space policy for NASA in the years since 1970.

Richard M. Nixon, "Statement About the Future of the United States Space Program," March 7, 1970

Over the last decade, the principal goal of our Nation's space program has been the moon. By the end of that decade men from our planet had traveled to the moon on four occasions and twice they had walked on its surface. With these unforgettable experiences, we have gained a new perspective of ourselves and our world.

I believe these accomplishments should help us gain a new perspective of our space program as well. Having completed that long stride into the future which has been our objective for the past decade, we must now define new goals which make sense for the seventies. We must build on the successes of the past, always reaching out for new achievements. But we must also recognize that many critical problems here on this planet make high priority demands on our attention and our resources. By no means

should we allow our space program to stagnate. But—with the entire future and the entire universe before us—we should not try to do everything at once. Our approach to space must continue to be bold—but it must also be balanced.

When this administration came into office, there were no clear, comprehensive plans for our space program after the first Apollo landing. To help remedy this situation, I established in February of 1969 a space task group, headed by the Vice President, to study possibilities for the future of that program. Their report was presented to me in September. After reviewing that report and considering our national priorities, I have reached a number of conclusions concerning the future pace and direction of the Nation's space efforts. The budget recommendations which I have sent to the Congress for fiscal year 1971 are based on these conclusions.

THREE GENERAL PURPOSES

In my judgment, three general purpose should guide our space program.

One purpose is exploration. From time immemorial, man has insisted on venturing into the unknown despite his inability to predict precisely the value of any given exploration. He has been willing to take risks, willing to be surprised, willing to adapt to new experiences. Man has come to feel that such quests are worthwhile in and of themselves—for they represent one way in which he expands his vision and expresses the human spirit. A great nation must always be an exploring nation if it wishes to remain great.

A second purpose of our space program is scientific knowledge—a greater systematic understanding about ourselves and our universe. With each of our space ventures, man's total information about nature has been dramatically expanded; the human race was able to learn more about the Moon and Mars in a few hours last summer than had been learned in all the centuries that had gone before. The people who perform this important work are not only those who walk in space suits while millions watch or those who launch powerful rockets in a burst of flame. Much of our scientific progress comes in laboratories and offices, where

dedicated, inquiring men and women decipher new facts and add them to old ones in ways which reveal new truths. The abilities of these scientists constitute one of our most valuable national resources. I believe that our space program should help these people in their work and should be attentive to their suggestions.

A third purpose of the United States space effort is that of practical application—turning the lessons we learn in space to the early benefit of life on earth. Examples of such lessons are manifold; they range from new medical insights to new methods of communication, from better weather forecasts to new management techniques and new ways of providing energy. But these lessons will not apply themselves; we must make a concerted effort to see that the results of our space research are used to the maximum advantage of the human community.

A CONTINUING PROCESS

We must see our space effort, then, not only as an adventure of today but also as an investment in tomorrow. We did not go to the moon merely for the sport of it. To be sure, those undertakings have provided an exciting adventure for all mankind and we are proud that it was our Nation that met this challenge. But the most important thing about man's first footsteps on the moon is what they promise for the future.

We must realize that space activities will be a part of our lives for the rest of time. We must think of them as part of a continuing process—one which will go on day in and day out, year in and year out—and not as a series of separate leaps, each requiring a massive concentration of energy and will and accomplished on a crash timetable. Our space program should not be planned in a rigid manner, decade by decade, but on a continuing flexible basis, one which takes into account our changing needs and our expanding knowledge.

We must also realize that space expenditures must take their proper place within a rigorous system of national priorities. What we do in space from here on in must become a normal and regular part of our national life and must therefore be planned in conjunction with all of the other undertakings which are also important to us.

As we enter a new decade, we are conscious of the fact that man is also entering a new historic era. For the first time, he has reached beyond his planet; for the rest of time, we will think of ourselves as men from the planet earth. It is my hope that as we go forward with our space program, we can plan and work in a way which makes us proud both of the planet from which we come and of our ability to travel beyond it.

Discouraged by warnings of constrained budgets from Nixon and the Office of Management and Budget, Thomas Paine left NASA in September 1970, and his deputy, George Low, was designated acting administrator. The first program that NASA had hoped to gain post-Apollo approval to develop was a twelve-person space station launched by the large rocket developed for Apollo, the Saturn V. In addition, NASA had proposed developing the space shuttle as a fully reusable Earth-to-orbit launch vehicle to carry crew and supplies to the space station and to carry out other space operations. When the Nixon administration as part of its decision to reduce the NASA budget ended Saturn V production and refused to approve the space station, NASA in the fall of 1970 revisited its plans. The space agency turned its attention to gaining White House approval for the space shuttle as NASA's centerpiece post-Apollo program. To do so, NASA had to present a new rationale for the shuttle, as a cost-effective way of launching all U.S. civilian and national security payloads and providing new capabilities for space operations. It did not make sense to develop the space station without a shuttle, but it was possible to create a rationale for developing a space shuttle without tying it to the space station.

These two letters from George Low reflect a fundamental reshaping of the post-Apollo human spaceflight program. NASA also knew that President Nixon wanted to continue U.S. human spaceflight. In his October 28 letter to OMB's Caspar Weinberger, Low pointed out that by approving shuttle development, the United States could have a continuing human spaceflight program without setting a major new space goal. This proved to be a winning argument.

Letter from George Low (Acting NASA Administrator), to George Shultz (Director, Office of Management and Budget), September 30, 1970

Dear Mr. Shultz:

The purpose of this letter is to transmit the FY 1972 budget recommendations for the National Aeronautics and Space Administration and to set forth the major considerations which influenced their development.

5. In recognition of current and longer term fiscal restraints, we have curtailed existing FY 1971 programs and adjusted the time-phasing of the future programs to avoid commitments to excessive funding levels in 1972 and future years. Our recent decisions to cancel two Apollo flights result in substantially lower expenditures than previously projected through FY 1974. Our decision not to proceed with the simultaneous development of the space station and space shuttle, and to defer until later in the 1970's the actual development of the post-Skylab space station, removes one of the principal causes of an unacceptable peaking of the NASA budget at over \$5 billion in the middle 1970's. As a result of these and other actions, we are now able to present and recommend an FY 1972 program which enables us to move toward each of the objectives in the President's statement without committing the nation to an annual NASA budget level in excess of \$4 billion.
6. The key element in our program for the 1970's is the space shuttle. It supports the last four of the President's six objectives. We must start this development now to lay the foundations for the nation's future space program, and to bring about the major economies in later years. This can be done within the fiscal limits previously discussed.

3. Reduce substantially the cost of space operations—The space shuttle will be used for manned and man-tended experiments and to place unmanned scientific, weather, earth resources, and other satellites in earth orbit and bring them

back to earth for repair and reuse. In the future the space shuttle will also transport men, supplies, and scientific equipment to and from space stations. Through reusability, the space shuttle system will have a recurring operating cost per mission substantially lower than the cost of current systems. Of even greater significance from a cost standpoint, major reductions will be possible in the cost of scientific and applications programs because of the relaxation of size and weight constraints and the capability for recovery, repair and reuse of payloads.

We will be ready to proceed in FY 1972 on a realistic schedule with detailed design and initial development efforts which we estimate will lead to the first horizontal test flights in 1975. The first vertical flight would take place in 1977 with a manned orbital flight test possible by the end of that year, leading to an initial operating capability in the 1978/1979 time period.

We place the highest importance on proceeding expeditiously with this program. Not only is it important to minimize the gap in U.S. manned space flights following Skylab in 1972-73, but economic analyses, which strongly support the development of the space shuttle on a cost-effective basis, show that a delay of a year in its availability development could increase the total cost of the U.S. space program by as much as \$2 billion.

4. Extend man's capability to live and work in space—The Skylab project, now in advanced stages of development, which requires substantial funding in FY 1972, is our only present flight program directed at this goal. Skylab will extend man's exposure to the space environment to 56 days, will perform an important manned solar astronomy experiment, and will extend our earth resources experiments beyond those carried out in the unmanned ERTS [Earth Resources Technology Satellite] program. After its launch in late 1972 and three revisit missions through the first half of 1973, no further manned missions using Apollo hardware are planned. We have made a major decision to defer development of a space station or "Skylab II" to a later time and to orient the space station studies we will continue in

FY 1972 toward modular systems that can be launched as well as serviced by the space shuttle.

Letter from George Low (Acting NASA Administrator), to Caspar Weinberger (Deputy Director, Office of Management and Budget), October 28, 1970

Dear Mr. Weinberger:

The discussions that NASA has had during the past weeks with OMB have covered, I believe, most of the matters of concern to us with regard to our FY 1972 budget. However, there are a few points—stated below in brief outline form—which I want to be especially sure are clearly understood and given careful consideration in the final decision process on the NASA FY 1972 budget.

2. Space Shuttle

- Our technical and management judgment is that we are progressing well on a realistic schedule of design studies and supporting technology development for the space shuttle, and will be ready to move in FY 1972 with detailed design and initial development as proposed in our FY 1972 recommendations.
- To support this schedule on an orderly, non-crash basis, the full amount (\$190 million) of FY 1972 estimates for the shuttle is required.
- The shuttle is clearly the key to effective and economical use of space in the future:
 - It will provide a single improved and more economical system for unmanned and manned missions of the future.
 - Its development can be justified as a versatile and economical system for placing unmanned civil and military satellites in orbit, entirely apart from its role in conducting or supporting manned missions.
- A clear go-ahead to the space shuttle system should be given at this time because:

- The U.S. should not have a national posture which leaves the field of manned space flight to the USSR after Skylab.
- With the shuttle, the U.S. can have a continuing program of manned space flight without maintaining, after 1973, the expensive base required for Apollo/Skylab operations and without a commitment to a major new manned mission goal.
- An FY 1972 go-ahead will already leave a significant gap in U.S. manned space operations. Any delay will only widen this gap
- Proceeding with the shuttle now will be a significant step in halting further erosion of U.S. aerospace capability and in offsetting the adverse impacts of current space and defense cutbacks on the U.S. aerospace industry, both of which are matters of serious concern
- A clear sign that the U.S. is proceeding with the shuttle is needed to support our efforts to secure substantial international participation; without it prospects for European participation—now surprisingly good—will evaporate.

The forgoing points, together with those made in my letter of September 30, require, in my view, careful consideration. We will, of course, be glad to provide any further information you require.

In 1970, NASA did not obtain the shuttle approval it sought. That approval needed to come in 1971, the agency argued, or its human spaceflight program would fade away after the 1973 launch of the Apollo-derived Skylab space station. A new NASA administrator, James Fletcher, arrived in May 1971. That summer, NASA and the White House were locked in a bitter battle over the agency's future programs and budgets. At issue was whether NASA would receive approval to continue the Apollo program through the Apollo 16 and 17 missions and to begin developing the space shuttle. Fletcher and Low, who stayed on as deputy administrator after Fletcher's arrival, soon decided to abandon a fully reusable shuttle design as both too expensive

and technically too challenging. The question then became what shuttle design NASA would propose.

In the second half of 1971, NASA scrambled to select a design that maintained the original shuttle's capabilities but could be developed on a substantially lower budget. Leading the skepticism regarding the wisdom of going ahead with the shuttle was the OMB staff. But OMB deputy director Weinberger disagreed with his staff; he decided that budget cutting was going too far. He told President Nixon in this August 1971 memorandum that not approving the shuttle and canceling the two remaining Apollo missions would be a mistake. Nixon wrote on the memo, "I agree with Cap." This exchange between Weinberger and the president was not reported to the OMB staff, who continued through the rest of 1971 to oppose NASA's shuttle proposal and to propose further reductions in the NASA budget.

Caspar W. Weinberger (Deputy Director, Office of Management and Budget), via George P. Shultz, Memorandum for the President, "Future of NASA," August 12, 1971

Present tentative plans call for major reductions or change in NASA, by eliminating the last two Apollo flights (16 and 17), and eliminating or sharply reducing the balance of the Manned Space Program (Skylab and Space Shuttle) and many remaining NASA programs.

I believe this would be a mistake.

1. The real reason for sharp reductions in the NASA budget is that NASA is entirely in the 28% of the budget that is controllable. In short we cut it because it is cuttable, not because it is doing a bad job or an unnecessary one.
2. We are being driven, by the uncontrollable items, to spend more and more on programs that offer no real hope for the future: Model Cities, OEO, Welfare, interest on National Debt, unemployment compensation, Medicare, etc. Of course, some of these have to be continued, in one form or another, but essentially they are programs, not of our choice, designed to repair mistakes of the past, not of our making.
3. We do need to reduce the budget, in my opinion, but we should not make all our reduction decisions on the basis of

what is reducible, rather than on the merits of individual programs.

4. There is real merit to the future of NASA, and to its proposed programs. The Space Shuttle and NERVA [an experimental nuclear-powered rocket engine] particularly offer the opportunity, among other things, to secure substantial scientific fall-out for the civilian economy at the same time that large numbers of valuable (and hard-to-employ-elsewhere) scientists and technicians are kept at work on projects that increase our knowledge of space, our ability to develop for lower cost space exploration, travel, and to secure, through NERVA, twice the existing propulsion efficiency of our rockets.

It is very difficult to re-assemble the NASA teams should it be decided later, after major stoppages, to re-start some of the long-range programs.

5. Recent Apollo flights have been very successful from all points of view. Most important is the fact that they give the American people a much needed lift in spirit, (and the people of the world an equally needed look at American superiority). Announcement now, or very shortly, that we were cancelling Apollo 16 and 17 (an announcement we would have to make very soon if any real savings are to be realized) would have a very bad effect, coming so soon after Apollo 15's triumph. It would be confirming, in some respects, a belief that I fear is gaining credence at home and abroad: That our best years are behind us, that we are turning inward, reducing our defense commitments, and voluntarily starting to give up our super-power status, and our desire to maintain world superiority.

America should be able to afford something besides increased welfare, programs to repair our cities, or Appalachian relief and the like.

6. I do not propose that we necessarily fund all NASA seeks—only that we couple any announcement to that effect with announcements that we are going to fund space shuttles, NERVA, or other major, future NASA activities . . .

Throughout the fall of 1971 suggestions for alternative shuttle designs emerged from NASA, its contractors, an advisory panel set

Step by presidential science adviser Ed David, and even from the contractor, Mathematica, that OMB had forced NASA to hire to carry out an analysis of shuttle economics. By November, OMB had developed, with the help of aerospace industry contractors it could not identify, a preferred shuttle design, a small vehicle with limited capabilities intended to test various aspects of shuttle reliability and capabilities. NASA settled on a larger, "full capability" version of the shuttle, with a large disposable external fuel tank and strap-on boosters to provide the initial thrust to lift the shuttle orbiter off the launchpad. On November 22, as a final decision on the shuttle neared, NASA administrator Fletcher made a "best case" argument for the NASA design to the White House.

James C. Fletcher, "The Space Shuttle," November 22, 1971

This paper outlines NASA's case for proceeding with the space shuttle. The principal points are as follows:

1. The U.S. cannot forego manned space flight.
2. The space shuttle is the only meaningful new manned space program that can be accomplished on a modest budget.
3. The space shuttle is a necessary next step for the practical use of space. It will help
 - space science,
 - civilian space applications,
 - military space applications, and
 - the U.S. position in international competition and cooperation in space.
4. The cost and complexity of today's shuttle is *one-half* of what it was six months ago.
5. Starting the shuttle now will have significant positive effect on aerospace employment. Not starting would be a serious blow to both the morale and health of the Aerospace Industry.

THE U.S. CANNOT FOREGO MANNED SPACE FLIGHT

Man has worked hard to achieve—and has indeed achieved—the freedom of mobility on land, the freedom of sailing on his oceans, and the freedom of flying in the atmosphere.